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SARDAR PATEL UNIVERSITY
BACHELOR OF COMPUTER APPLICATION (SEMESTER – V) EXAMINATIONS – 2025
US05CBCA53: SOFTWARE ENGINEERING

Date: 07/10/2025(Tuesday)
Time: 10.00 AM to 01.00 PM

Total: 70 Marks

Q-1 Multiple Choice Questions.

[10]

1. _____ is the collection of computer programs, procedures and data.
A. Hardware
B. Software
C. Network
D. Engineering
2. _____ is the simplest and most widely used software development model.
A. Spiral
B. Prototype
C. Iterative enhancement
D. Waterfall
3. A high quality SRS reduces the development _____.
A. Time
B. Customer requirements
C. Cost
D. Quality
4. COCOMO stands for:
A. Construction Cost Model
B. Constructive Cost Model
C. Constructive Code Model
D. Calculated Cost Model
5. KDLOC means _____.
A. Kilogram Developed Line of Code
B. Kilogram Delivered Local Code
C. Thousands Delivered Local Code
D. Thousands Delivered Line of Code
6. Structured design methodology tries to reduce _____.
A. Cost
B. Time
C. Cohesion
D. Coupling
7. Number of subordinates associated with given module is known as _____.
A. Fan-out
B. Fan-in
C. Dependency
D. Module
8. Structured programming is often called _____ programming.
A. goto-less
B. object oriented
C. procedural
D. None of these
9. Which static method is used for verify the programs?
A. Review
B. Automated cross checking
C. Code reading
D. None of these
10. A failure is produced only when there is a _____ in the system.
A. error
B. bug
C. fault
D. problem

Q-2 Short Questions. (Attempt any Ten).

[20]

1. Define: Coupling and Cohesion.
2. List the levels of Cohesion.
3. List down characteristics of software process.
4. What are the limitations of Waterfall model?
5. Explain Design constraints as a component of SRS.
6. List down at least 8 variables of COCOMO model.
7. Explain Partitioning.
8. List at least 6 general rules of programming style.
9. Draw the diagram of levels of testing.
10. Write a short note on Earned Value method.
11. Write the goal of coding.
12. Explain in brief Design walkthrough.

Q-3 Explain phases of software development.

10

OR

Q-3 Explain waterfall model.

10

Q-4 Explain COCOMO Model.

10

OR

Q-4

[A] What is SRS? Explain needs of SRS.

05

[B] Explain general characteristics of SRS.

05

Q-5 What is system design? Explain how coupling and cohesion used.

10

OR

Q-5

[A] Write a short note on Coupling.

05

[B] What is design specification? Explain factors of it.

05

Q-6

[A] Explain the Top-Down and Bottom-Up approach in coding.

05

[B] What is verification process in coding phase and explain Code reading method.

05

OR

Q-6 What is testing? Explain levels of testing using diagram.

10

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